

Cover Letter

Dear Editors,

Please consider our manuscript, "Benchmarking Gaussian Splatting Under Sparse and Degraded Views", for publication in Scientific Reports.

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The manuscript fits Scientific Reports because it presents a technically sound and reproducible computational benchmark using public 3D reconstruction scenes, controlled sparse-view settings, repeated random seeds, and explicit degradation tests. The work is not a demonstration-only system; it provides measured evidence, limitations, audit files, and reproducibility materials.

The novelty is a compact benchmark package that combines sparse-view 3D Gaussian Splatting, quality and diversity-based view selection, degradation sensitivity reported relative to clean baselines, foundation-model pose proxy results, and a limited InstantSplat rendering study. The conclusions are cautious: quality-only selection is weak in the evaluated scenes, view diversity matters, and random/uniform baselines remain strong comparators.

The cleaned code and generated result materials are publicly available at <https://github.com/XessX/sparse3d-gaussian-splatting-benchmark> and archived at <https://doi.org/10.5281/zenodo.20478630>. Raw public dataset images are not redistributed.

We confirm that the manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors approve this submission.

Suggested reviewers: [MANUAL: suggested reviewer name, institution, email]; [MANUAL: suggested reviewer name, institution, email]

Excluded reviewers: None declared.

Prior editorial discussion: No prior discussion with a Scientific Reports editorial board member has occurred, unless updated before submission.

Sincerely,

Al Jubair Hossain